radiation background. They may not constitute a hazard to the workers, but may ruin the scientific experiments if not cleaned up immediately. In all work with radionuclides, radioactive waste is produced. It is common practice to collect all such waste in special containers, and to dispose of it according to national rules. For short lived radionuclides of low hazard and low levels of radioactivity (e.g. as in ^{14}C-work), it is common practice to dispose of such waste by normal flushing to the sewer with several liters of tap water if such procedures are permitted by the national radiation protection organizations.

15.14. PROTECTIVE MEASURES FOR RADIOCHEMICAL LABORATORY WORK

Three basic principles are recommended for keeping radiation exposure to a minimum: shielding, control, and distance. If a radiochemical laboratory is designed properly and the work performed in such a manner that the general background contamination is sufficiently low to do valid low level tracer experiments, then the health aspects of radiation control are satisfied. The main principles for work with radioactive substances are indicated, but in each notion, special rules may apply.

15.14.1. Tracer work with moderate β-γ-levels

Ordinary chemical laboratories may be used for radiochemical work at low levels of short lived β-γ radionuclides (e.g. half-lives <14 d and activity levels <10 kBq). However, it is recommended that a special room be used for radioactive work. In the design of such a laboratory it is important that airborne contamination be prevented from spreading to counting rooms and to offices. Therefore, a pressure difference between the laboratories and the other areas is desirable. Thus, the air velocity in the fume hoods should never be below 0.25 m s^{-1}, and 0.5 m s^{-1} is recommended. With such a flow velocity, radioactive dust and fumes are retained in the hood and removed through the vents. The fume hoods should have filters for collecting radioactive particulates.

To minimize the possibility of ingestion as well as the chances of ruining experiments through accidental contamination, limiting the radioactive work to a minimum area is essential. There should be no radioactivity except in the immediate working area, and upon completion of the particular experiment, all activity should be removed and the area cleaned (*decontaminated*) if necessary. For low levels, this means working in a good hood with easily cleaned nonporous surfaces.

One operation which commonly results in contamination involves evaporation of a solution to dryness either on a hot plate or under a heat lamp. Although the percentage of the sample carried away by spray may be very small, it may result in appreciable amounts of activity being spread around the area of the evaporation. Consequently, all evaporation

should be performed in a hood and the vicinity should be protected from the active spray by a covering of absorbent paper.

For nuclides which emit only β-particles, the glass walls of the container may provide sufficient shielding. Sheets of glass or plastic (such as lucite) are commonly used to shield exposed solid samples.

For work with higher levels of β-γ-emitters of longer half-lives, special radiochemical laboratories should be used. The working surfaces and floors should be even, nonporous, and with a minimum of seams. Surfaces of plastic material, stainless steel, and artificial stone are acceptable bench materials. It is recommended that fume hood and benches be covered with an absorbing material such as absorbing paper, and that experiments be conducted when possible in trays of stainless steel or plastic. Such arrangements ensure that the radioactive material does not contaminate a larger area if a spill occurs. Radionuclide workers should wear surgical gloves and laboratory coats in the laboratory. If there is any danger of splashing, plastic face-hoods are recommended. It is extremely important that oral contamination be avoided. Beta-radiation from radioactive sources has ranges which rarely exceed 1 g cm^{-2}. Consequently, in a laboratory in which the level of β-emission is less than 1000 MBq, protection from the radiation can be achieved with a 1 cm plexiglass shield.

15.14.2. α-laboratories

For α-emitters the main hazard is internal, not external, to the body. For moderate activity levels a *glove box* under slightly reduced pressure provides a simple and convenient closed chemical laboratory. Samples of the actinide elements usually have high specific radioactivities. Therefore, special care must be exercised in working with them. Because these α- emitters usually are associated with weak γ-radiation or X- rays, chemical work with these elements must be conducted in more advanced gloveboxes (Fig. 15.16), which are kept at a pressure slightly below the surrounding atmosphere by circulation of pure air or inert gas through the box. The boxes should have alarm systems for monitoring hazards such as interrupted water circulation, electrical short circuits, oxygen in the inert gas, heat, etc. In Figure 15.16, the control panel for these protective arrangements is shown above the box. In large alpha-box laboratories, one of the main hazards is radioactive dust. All room surfaces should, therefore, be made with as few seams and sharp corners as possible; particularly the floor must be of high quality. Electric power, water, waste, ventilation, etc., should be connected to piping in the ceiling. The air into the laboratory must pass through filters as must the air exiting the laboratory. The exit air should be monitored for α-activity. Entering and leaving the laboratory should be through airlocks, and the hands and feet must be monitored for activity on exit. All these protective measures make α-laboratories quite expensive, but smaller laboratories working with lower levels of α-activity can be constructed in simpler fashion for correspondingly less cost.

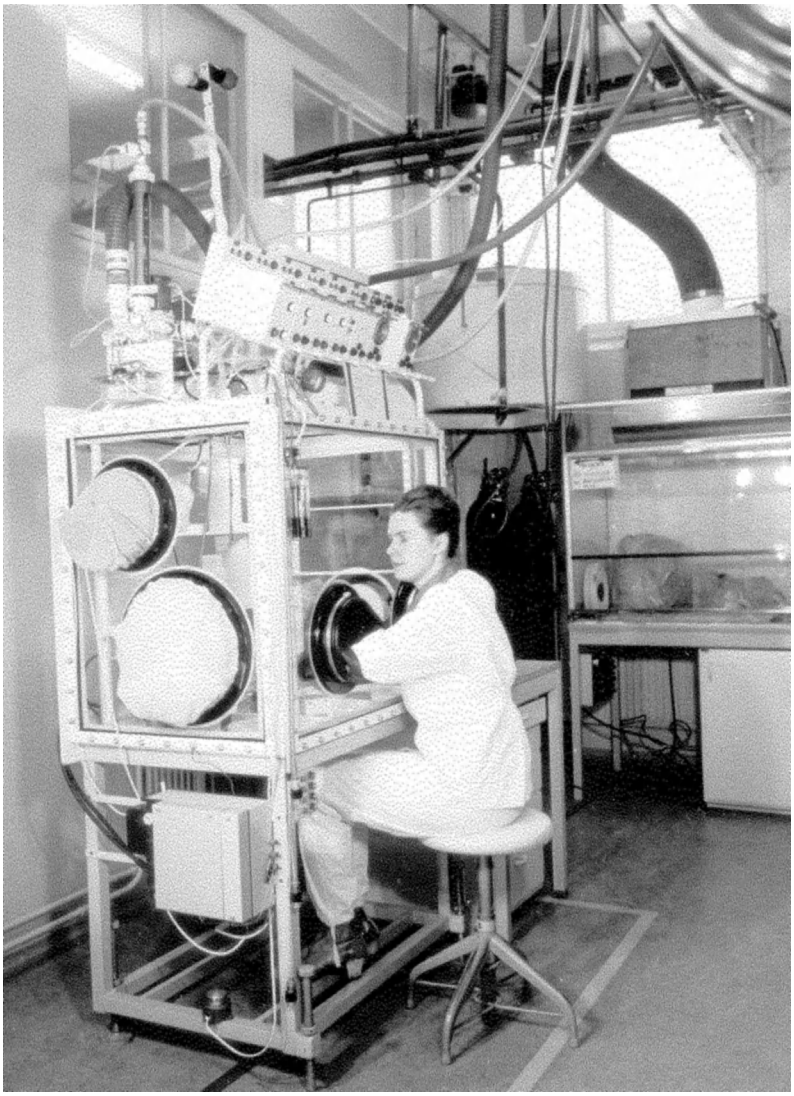


Figure 15.16 Glove-box for plutonium experiments. A radiochemical fume hood with filter is seen in the background. Note that all connections are in the ceiling.

15.14.3. High level β-γ-emitters

Since the intensity decreases as the inverse square of the distance, maintaining maximum distance (by use, when necessary, of remote control apparatus such as tongs) when working with moderate or high levels of activity reduces the exposure appreciably.

High levels of γ, >1000 MBq, require shielding with layers of concrete, water, steel, or lead, and the operations must be carried out by remote control. The eyes can be protected by the use of lead glass windows of high density, periscopes, mirrors, etc., see Figure 15.17.

Commonly, special shielded cells are used. These cells are sealed from the atmosphere and kept at a pressure lower than that for the working personnel. The smallest cells usually have lead walls which sometimes reach to the ceiling; by this arrangement, no scattered radiation reaches the working personnel. Experiments are carried out with the use of tongs passing through the lead walls or, for the thicker cells, with manipulators reaching above or through the walls. In cells for very high activities (≥10 TBq, or >1000 Ci), these manipulators are guided electrically or mechanically (*master-slave manipulators*). All movements of the operator are copied exactly by the slave hands inside the cell. For very complicated work and extremely high radioactivity, robots have been

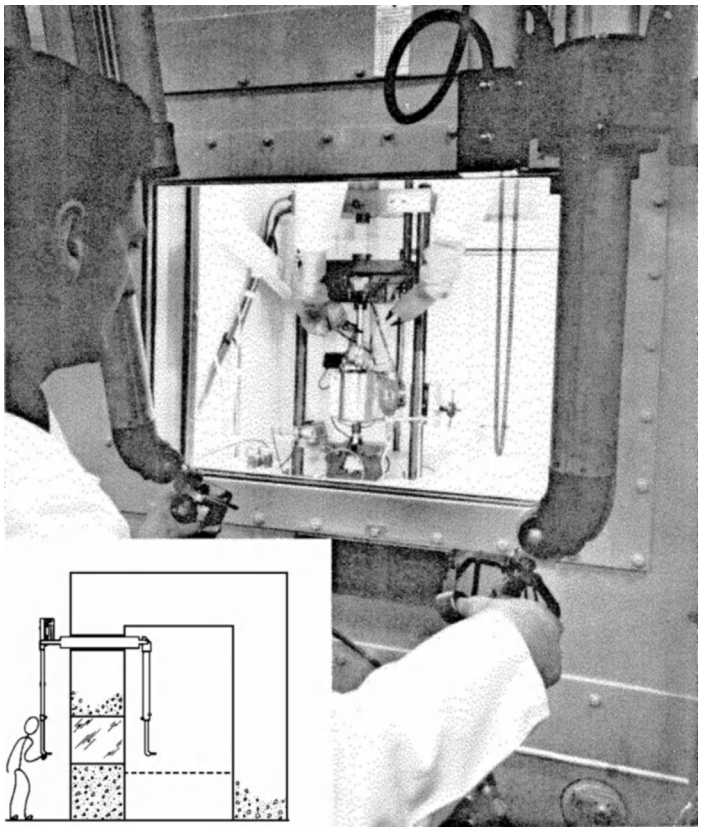


Figure 15.17 High-active β-γ-work with master-slave manipulators. The window is filled with high density $ZnBr_2$ solution.

developed which can be guided to repair such items as unshielded nuclear reactors, radionuclide equipment, heavily contaminated radiochemical apparatus, etc.

15.15. CONTROL OF RADIATION PROTECTION MEASURES

In larger organizations, the control of radiation hazards is the responsibility of specialists known as *health physicists* or, sometimes, *health chemists*. Their main duty is to ensure that work is carried out without hazard to the health of the people involved.

The protection follows three stages: prevention, supervision, and after-control. Preventive measures include use of fume hoods, α-boxes, radiation shielding, tongs, etc., as discussed above. The supervision stage involves the use of radiation instruments to monitor the radiation level (see Ch. 9). Small TLD, film or pocket pen dosimeters are used for individual monitoring (§8.9). For spills and contamination of hands, shoes, etc., special contamination instruments (counters) are used which are more sensitive than the monitoring dose instruments.

Contamination in the laboratory must be avoided. This is controlled by smear tests; i.e., a filter paper is wiped over the surface and the paper is checked with a suitable instrument. In a so-called "clean area", the fixed contamination should not exceed 2 Bq for α, and 4 Bq for β-γ on a surface of 100 cm^2. For an "active area" the rules are a maximum of 20 Bq for α per 100 cm^2, and 0.01 mGy h^{-1} from β—γ at a distance of 2 cm from the surface. Radioactive aerosols are monitored by air samplers in which a certain amount of air is drawn through a fine filter paper after which the paper activity is measured.

The after-control usually consists of checking personal dosimeters and a medical examination. Depending on the kind of work, the dosimeters are checked from twice a week to once a month. A medical examination is given once or several times a year, depending on the work conditions. In danger of inhalation or ingestion of α- or soft β-emitters, urine samples are analyzed. Such analyses are very sensitive and much less than a kBq in the body is easily detected. For workers who handle hazardous amounts of α-emitters (e.g. plutonium in more than milligram amounts), urine samples should be taken regularly.

If necessary whole-body counts are also taken. *Whole-body counting* is carried out with the subject being surrounded by numerous scintillation or solid state detectors in a heavily shielded room. The natural body content of ^{40}K is easily detected and is a control of the efficiency of this technique.

15.16. EXERCISES

15.1. "Reference man" consists of 18%C, 66% H_2O, 0.2% of K per body weight. He may also have accumulated 10 pCi ^{226}Ra in the body; assume 0.3 decay for each of